

MEC V1.0 - Modular Embedded Controller

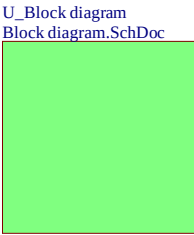
Table of contents

- Sheet 1: Project overview (this page)
- Sheet 2: Block diagram
- Sheet 3: Main project
 - Sheet 4: Power Supply (PWR)
 - Sheet 5: Digital Input (DIN)
 - Sheet 6: Digital Output (DOUT)
 - Sheet 7: Industrial Serial Comm (RS485)
 - Sheet 8: RF Transceiver Module (RF)
 - Sheet 9: Ethernet Interface (ETH)
 - Sheet 10: Microcontroller (MCU)

Uniformar librería (BOM)

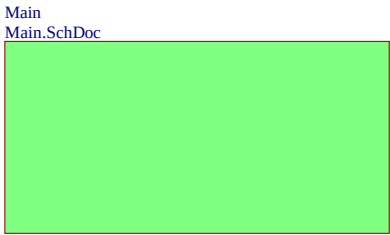
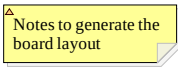
Uniformar librería Schematics

Uniformar librería Footprints según IPC



Legend

- General comment such function title, configurations, datasheets, ...
- Text to be added to silkscreen
- Design references
- Warning text
- DNP Do-not-place



Title: Project overview.SchDoc	
Project: MEC.PrjPcb	
Revision: 1.0	Sheet: 1 of 10
Size: A4	Date: 12/11/2025
Draw by: Ing. Lucas Kirschner	



1

2

3

4

A

A

B

B

C

C

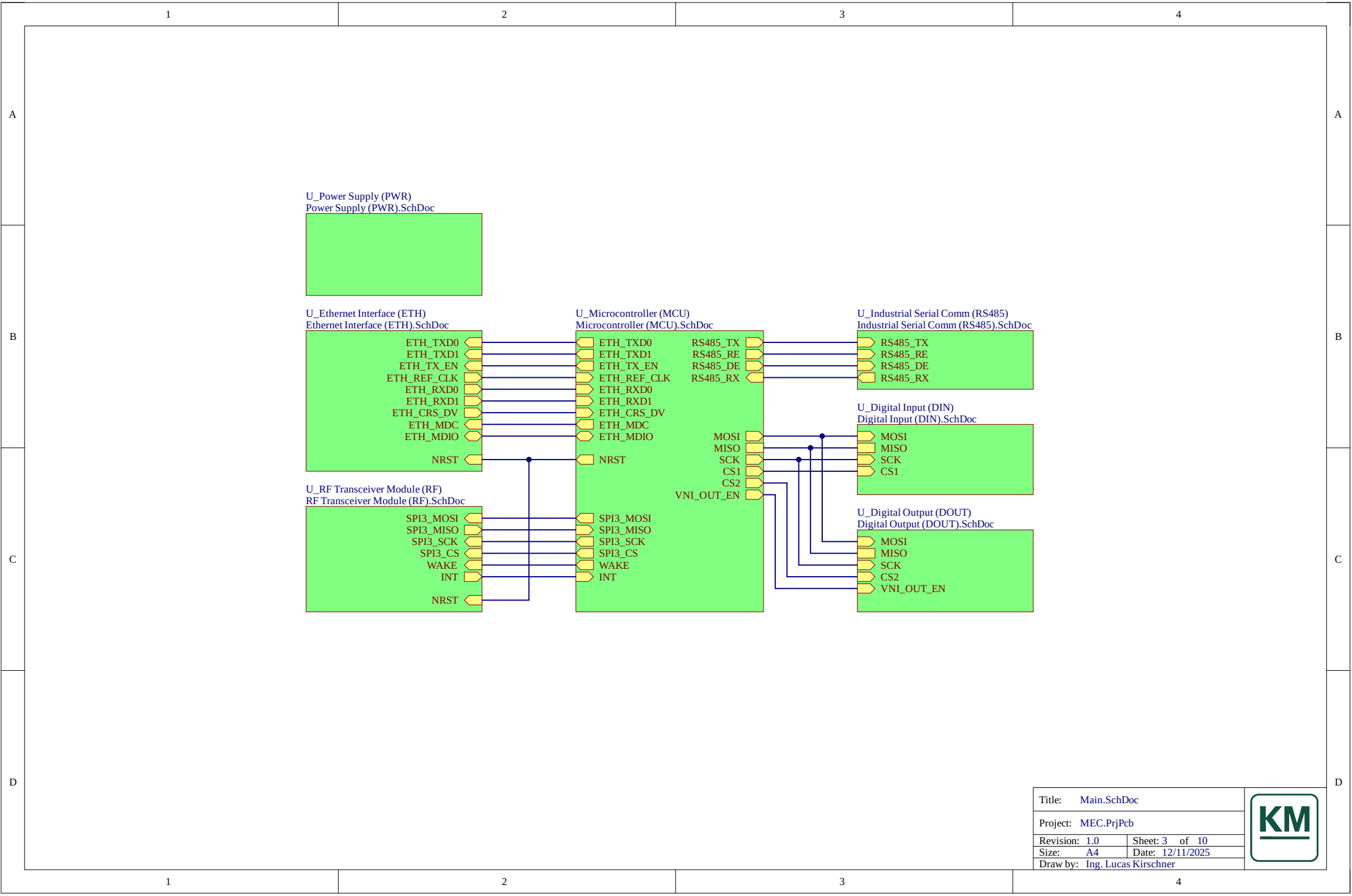
D

D

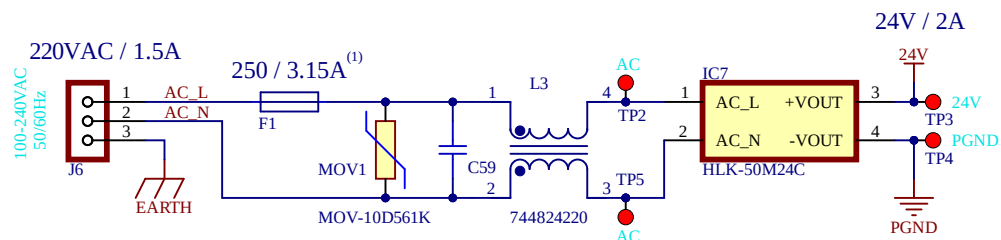
Reservado diagrama de bloques

Title: Block diagram.SchDoc	
Project: MEC.PrjPcb	
Version 1.0	Sheet: 2 of 10
Size: A4	Date: 12/11/2025
Draw by: Ing. Lucas Kirschner	

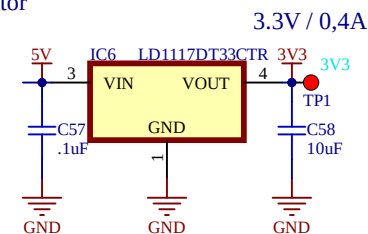




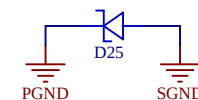
220VAC to 24VDC AC/DC converter



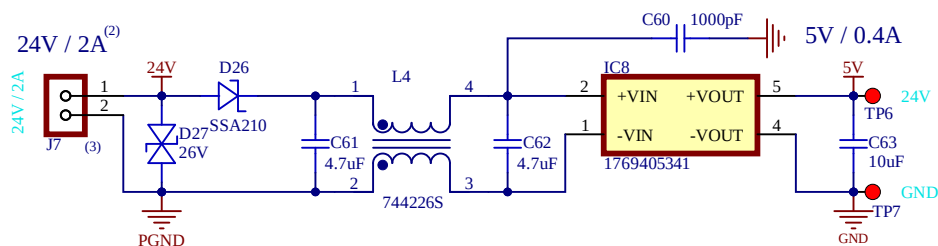
3V3 linear regulator



GND NET Tie



24V to 5V DC/DC isolated converter



References

- 1) Fusible de acción media para la entrada de red (AC). Modelo recomendado: Littelfuse 02323.15MXP.
- 2) Corriente máxima total del riel de 24 V: 2 A. Cuando el conector de 24 V actúa como salida (módulo HLK-50M24C instalado), este valor corresponde a la suma de todos los consumos del sistema. La corriente disponible en el conector disminuirá si las salidas digitales entregan potencia.
- 3) El conector de 24 V puede funcionar como entrada o salida, según la presencia o ausencia del módulo HLK-50M24C en el circuito. No aplicar 24 V externos si el módulo HLK-50M24C está instalado.

Title: Power Supply (PWR).SchDoc

Project: MEC.PrjPcb

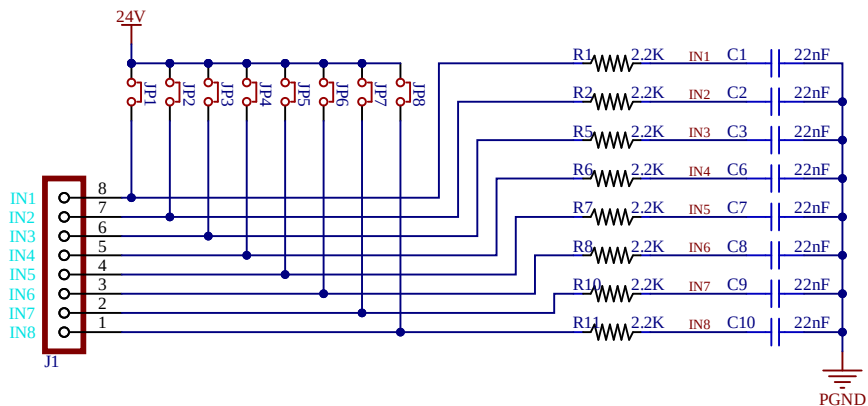
Revision: 1.0 Sheet: 4 of 10

Size: A4 Date: 12/11/2025

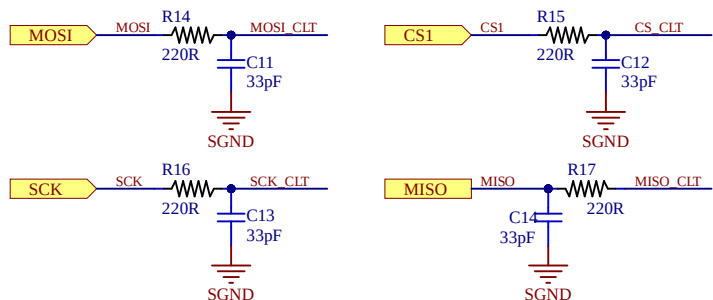
Draw by: Ing. Lucas Kirschner



Digital input connector



SPI signals

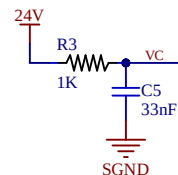


References

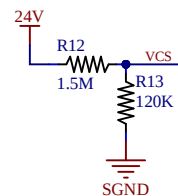
See <https://www.st.com/en/protections-and-emi-filters/sclt3-8bt8>
 See <https://www.st.com/en/evaluation-tools/x-nucleo-plc01a1>
 See <https://www.st.com/en/evaluation-tools/steval-ifp031v1>
 See <https://www.st.com/en/evaluation-tools/steval-ifp007v1>

High speed digital input current limiter

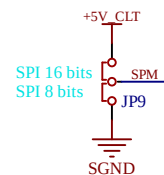
24 V sensor power supply



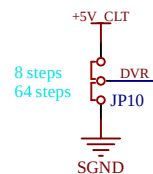
24 V sensor power supply sensing input



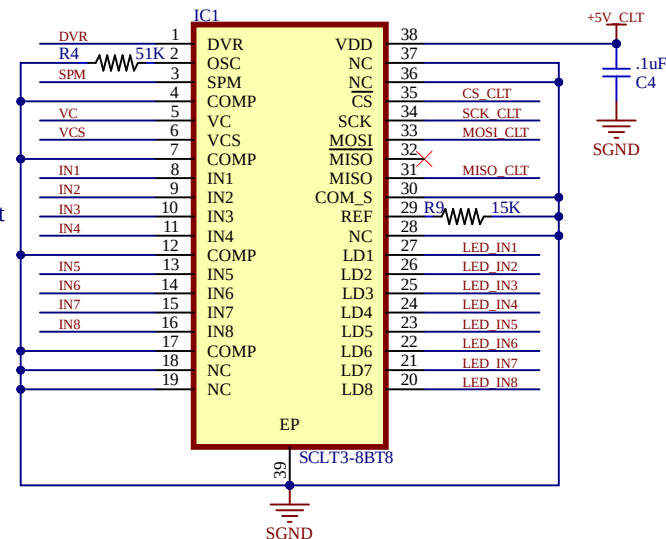
SPI 8/16-bit selection



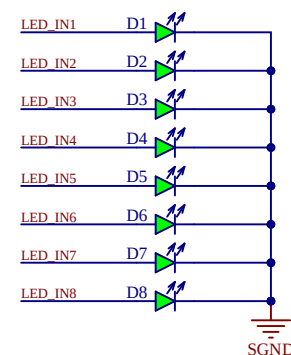
Divider ratio selector of the digital input filters



VDD max power supply: 12mA



LED input driver



Title: Digital Input (DIN).SchDoc

Project: MEC.PrjPcb

Revision: 1.0

Sheet: 5 of 10

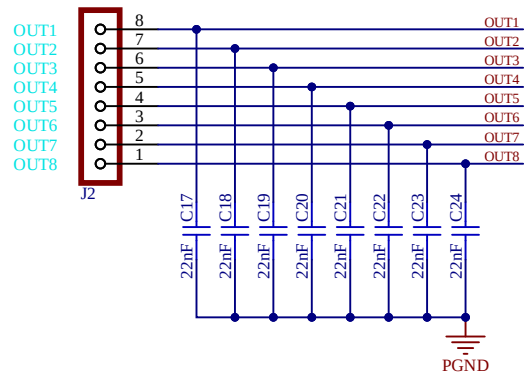
Size: A4

Date: 12/11/2025

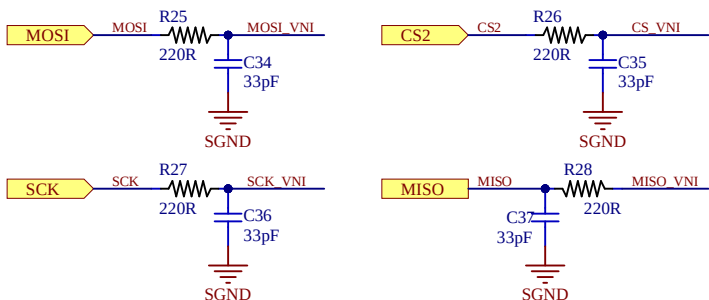
Draw by: Ing. Lucas Kirschner



Digital output connector



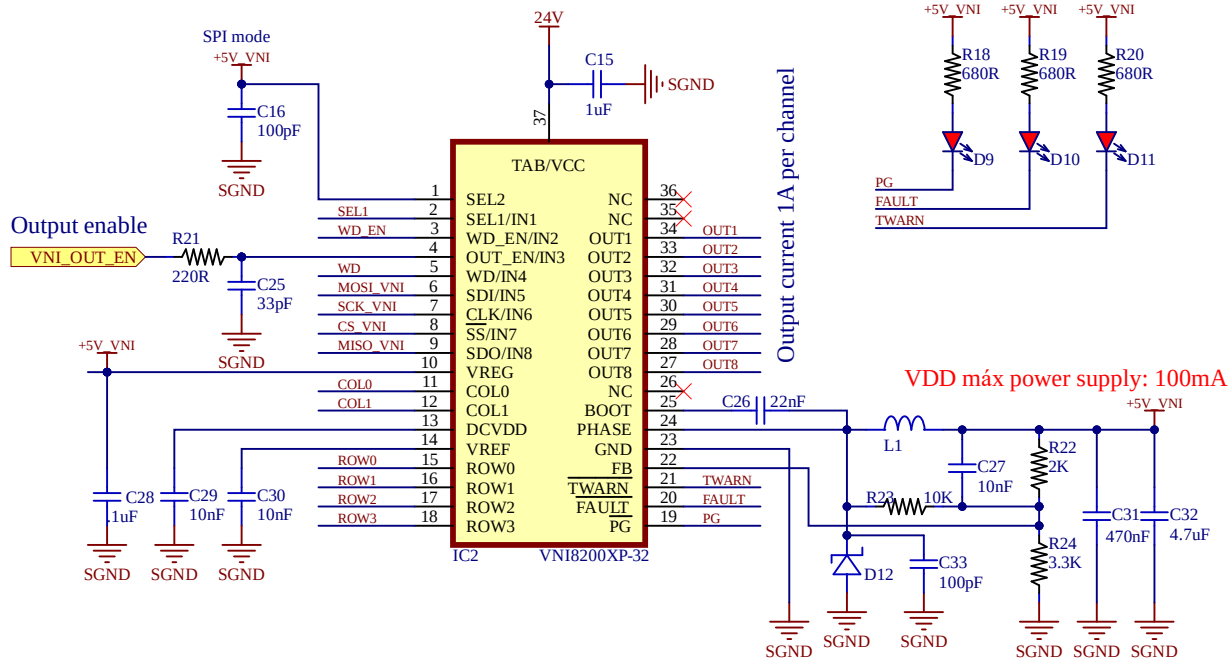
SPI signals



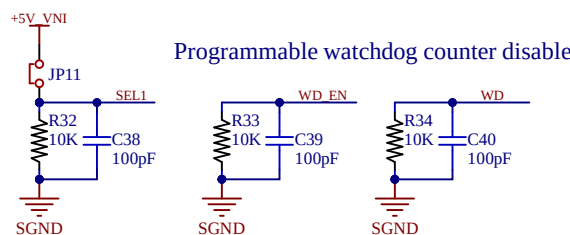
References

See <https://www.st.com/en/power-management/vni8200xp-32>
 See <https://www.st.com/en/evaluation-tools/x-nucleo-plc01a1>
 See <https://www.st.com/en/evaluation-tools/steval-ifp032v1>
 See <https://www.st.com/en/evaluation-tools/steval-ifp022v1>

Octal high-side smart power interface

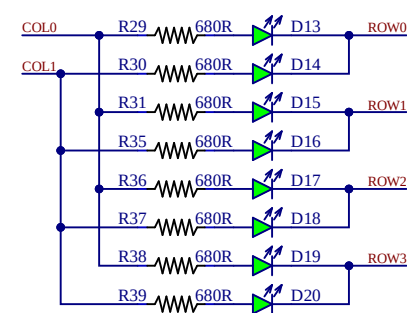


SPI 8/16-bit selection

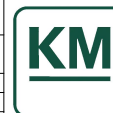


Programmable watchdog counter disable

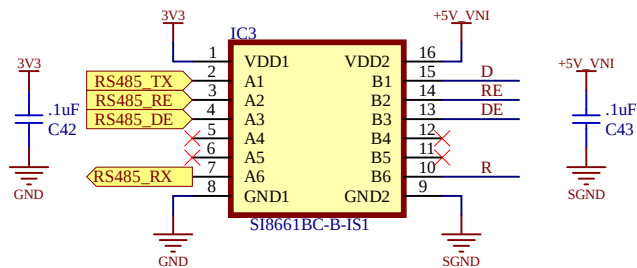
LED output driver



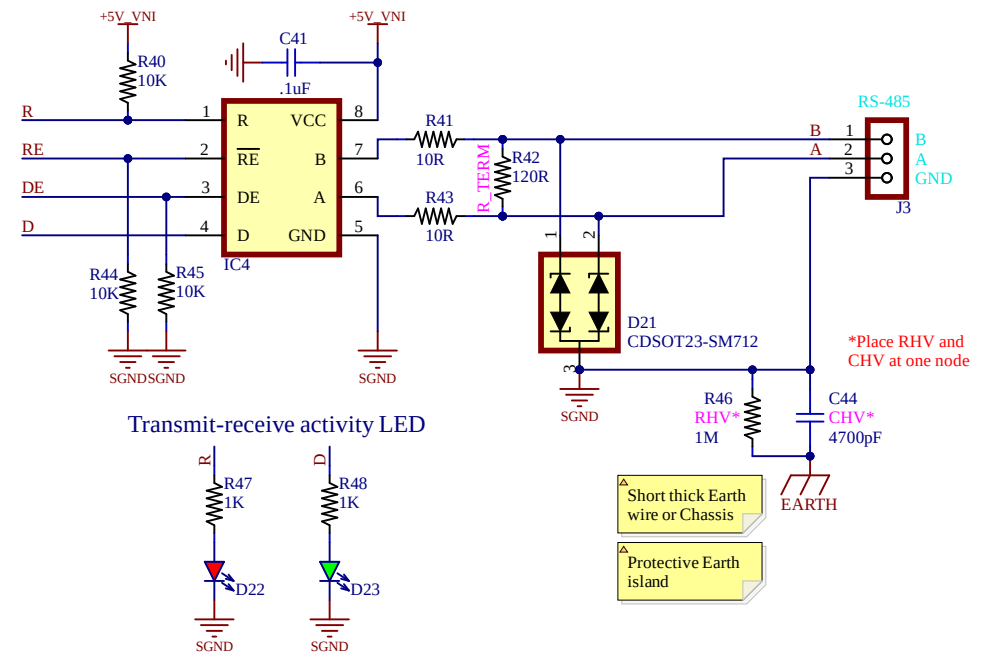
Title: Digital Output (DOU).SchDoc
 Project: MEC.PrjPcb
 Revision: 1.0 Sheet: 6 of 10
 Size: A4 Date: 12/11/2025
 Draw by: Ing. Lucas Kirschner



Digital isolator



RS485 interface



A: Driver output or receiver input (complementary to B)
B: Driver output or receiver input (complementary to A)
D: Driver data input
DE: Driver enable, active high
R: Receive data output
RE: Receiver enable, active low

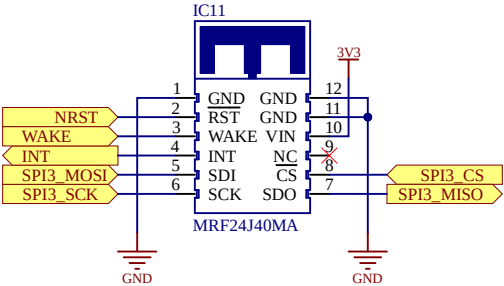
References

See <https://www.ti.com/product/SN65HVD82>
 See <https://www.st.com/en/evaluation-tools/x-nucleo-plc01a1>
 See <https://www.ti.com/tool/RS485-HF-DPLX-EVM>
 See <https://www.ti.com/lit/an/slla070d/slla070d.pdf>

Title: Industrial Serial Comm (RS485).SchDoc	
Project: MEC.PrjPeb	
Revision: 1.0	Sheet: 7 of 10
Size: A4	Date: 12/11/2025
Draw by: Ing. Lucas Kirschner	



RF Transceiver Module



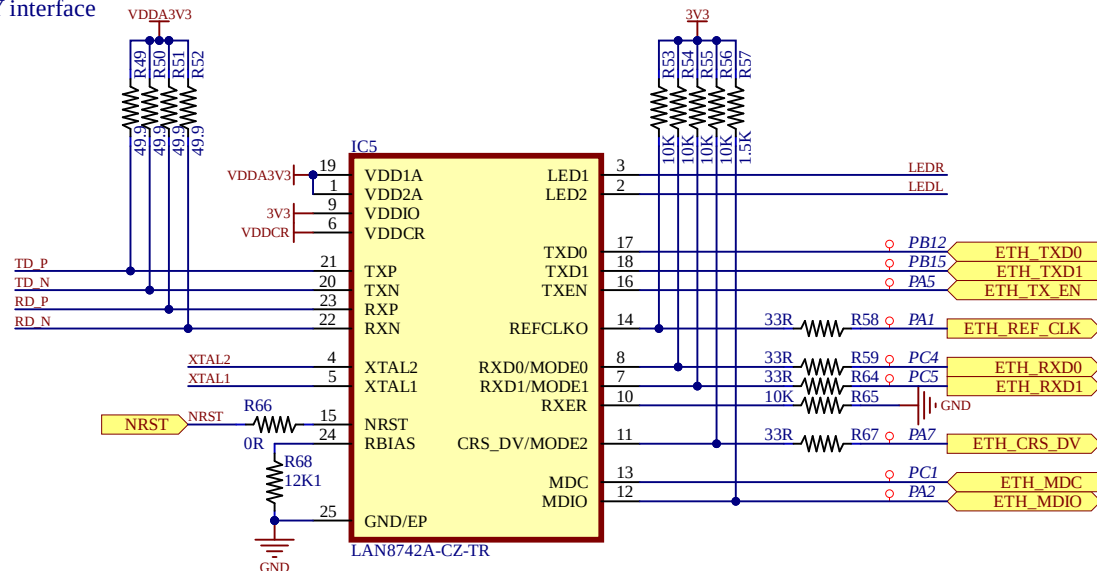
References

See <https://www.microchip.com/en-us/product/MRF24J40MA>

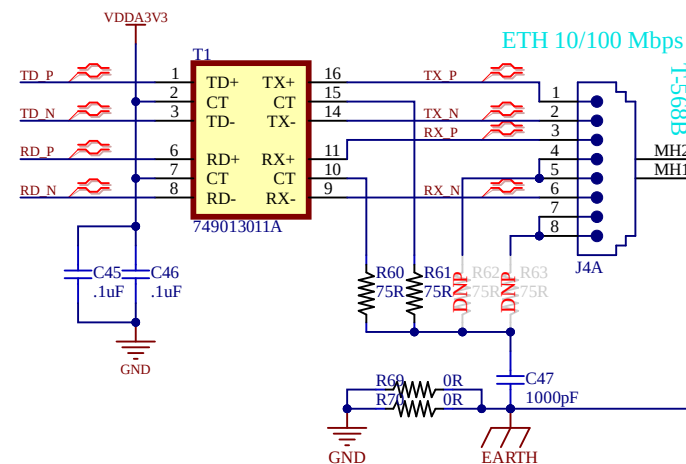
Title: RF Transceiver Module (RF).SchDoc	
Project: MEC.PrjPcb	
Revision: 1.0	Sheet: 8 of 10
Size: A4	Date: 12/11/2025
Draw by: Ing. Lucas Kirschner	



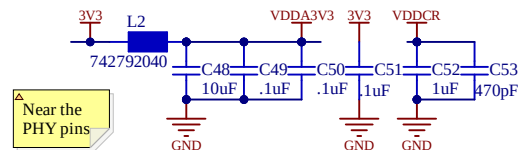
PHY interface



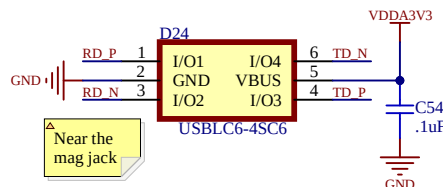
Ethernet transformer & connector



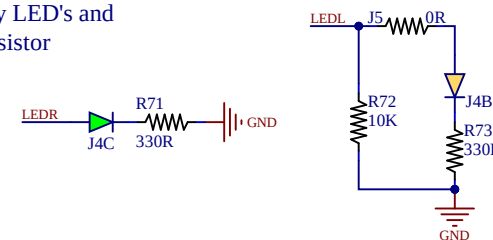
PHY decoupling capacitors



ESD protection



Activity LED's and strap resistor

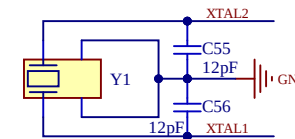


References

- Los resistores de pull up en los pines MODE0, MODE1 y MODE2 son resistores de strap que configuran al LAN8742 en el modo "all capable" con la autonegociación activada.
- LEDL cuenta con un resistor de strap que configura el pin REFCLKO como salida de reloj de referencia para la capa MAC, si se deja flotado o conectado a VDD2A, el pin indica interrupciones nINTSEL.

See <https://www.st.com/en/evaluation-tools/nucleo-h563zi>
See https://www.microchip.com/downloads/DS_LAN8742_00001989A.pdf

External crystal



Title: Ethernet Interface (ETH).SchDoc

Project: MEC.PrjPcb

Revision: 1.0 Sheet: 9 of 10

Size: A4 Date: 12/11/2025

Draw by: Ing. Lucas Kirschner



